The molecular evidence of *Babesia microti* infection in small mammals collected in Slovenia

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SUMMARY

In Europe, the zoonotic cycle of *Babesia microti* has not been determined so far. Recently, *B. microti* was detected in *Ixodes ricinus* ticks in Slovenia by using molecular methods. In order to investigate the mammalian hosts of *B. microti* in Slovenia we collected 261 small mammals representing 11 species. They were tested for the presence of babesial parasites with a PCR assay based on the nuclear small subunit rRNA gene (nss-rDNA). The bank vole (*Clethrionomys glareolus*) and yellow-necked mouse (*Apodemus flavicollis*) were infected with *B. microti*. The prevalence rate was 15·9% for *C. glareolus* and 11·8% for *A. flavicollis*. Nucleotide sequences of amplified portions of *B. microti* nss-rDNA from *C. glareolus* and *A. flavicollis* were indistinguishable from each other and identical with those previously described in *I. ricinus* ticks collected in Slovenia. The results of this study represent molecular evidence of *B. microti* in small mammals in Europe.

Key words: *Babesia microti*, *Clethrionomys glareolus*, *Apodemus flavicollis*, molecular analysis, Slovenia.

INTRODUCTION

Protozoa of the genus *Babesia* are some of the most ubiquitous and widespread blood parasites (Telford *et al*. 1993). *Babesia microti* is the main causative agent of human babesiosis, a tick-borne zoonosis increasingly recognized in the north-eastern and mid-western USA (Homer *et al*. 2000; Kjemtrup & Conrad, 2000). The life-cycle of *B. microti* in the USA has been well described: the white-footed mouse (*Peromyscus leucopus*) is the most important reservoir and the tick *Ixodes scapularis* is the main vector (Piesman & Spielman, 1980; Spielman *et al*. 1981; Telford & Spielman, 1993). *Microtus pennsylvanicus*, *M. montanus* and *M. ochrogaster* serve as minor reservoir hosts (Watkins *et al*. 1991; Burkot *et al*. 2000).

However, the zoonotic status of *B. microti* in Europe remains an open question. In different countries across Europe this parasite has been found in a variety of small mammals. The absence of indigenous cases of babesiosis caused by *B. microti* was explained by the selective feeding behaviour of the proposed vector *I. trianguliceps* (Kjemtrup & Conrad, 2000). Although *I. trianguliceps* was experimentally established as a competent vector of *B. microti*, this tick species does not bite humans (Randolph, 1995). Recently, *B. microti* has been detected in *I. ricinus* in Slovenia, Poland and Switzerland by means of molecular identification (Duh, Petrovec & Avsic-Zupanc, 2001; Skotarczak & Cichocka, 2001; Foppa *et al*. 2002). In transmission studies Gray *et al*. (2002) demonstrated that *I. ricinus* could serve as a vector of *B. microti* in Europe. *I. ricinus* is defined as an opportunistic tick species utilizing a wide range of vertebrates as hosts and it readily bites humans (Sonenshine, 1993).

In Slovenia, the reservoir of *B. microti* had not been investigated. Therefore we conducted this study to determine whether small mammals, captured in Slovenia, are infected with babesial parasites and also to assess the DNA sequence homologies between amplicons derived from small mammals and those previously obtained from *I. ricinus* ticks.

MATERIALS AND METHODS

Field collections

Small mammals were trapped between 1996 and 2000 in 2 ecologically different regions of Slovenia: the Subpannonian (UTM: WM77, WM88, WM96, XM08, XM09) and the Pre-alpine (UTM: VM41, VM51, VL49, VL58) region. The vegetation in these regions comprises predominantly broad-leaved forest, coniferous forest of boreal type and elements of the mountainous Mediterranean (Mrsic, 1997). Animals were captured in live traps, Eliot special and Sherman type, and sacrificed by ether hyperanaesthesia. To aid identification, animals were weighed and total head–body length, tail, ear and hind-foot lengths were measured. The blood was collected from the axillar artery and thin blood smears were immediately prepared.

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Table 1. Detection of *Babesia microti* in small mammals, captured in Slovenia, using PCR assays and subsequent sequence analysis

Species	Common name	No. tested	Percentage *B. microti* positive (no.)
Apodemus flavicollis	Yellow-necked mouse	127	**11·8** (15)
Clethrionomys glareolus	Bank vole	69	**15·9** (11)
Apodemus agrarius	Striped field mouse	35	0 (0)
Apodemus sylvaticus	Wood mouse	17	0 (0)
Microtus arvalis	Common vole	4	0 (0)
Microtus agrestis	Field vole	1	0 (0)
Neomys fodiens	Water shrew	1	0 (0)
Neomys anomalus	Miller's water shrew	1	0 (0)
Sorex araneus	Common shrew	3	0 (0)
Sorex minutum	Pigmy shrew	1	0 (0)
Mus musculus	Eastern house mouse	2	0 (0)

Microscopy

Thin blood smears were air-dried and stained with The Dade Diff-Quik Staining Set (Dade AG, Dudingen, Switzerland) which is a modification of the Pappenheim technique (Giemsa-May-Grünwald). Blood smears were examined by light microscopy at $\times 1000$.

Analysis of DNA sequences

DNA was extracted from a small piece (1 mm^3) of spleen from individual animals using the QIAamp DNA Tissue Kit (Qiagen, GmbH, Hilden, Germany). Homogenized samples were lysed by overnight incubation at 56 °C in the lysis buffer. Samples were applied to the QIAamp spin column following the manufacturer's instructions.

To detect babesial parasites, each sample was tested with primers PIRO-A and PIRO-B, which were designed to amplify the 407, 408 and 435 basepair (bp) fragments of the nuclear small subunit ribosomal RNA gene (nss-rDNA) of *B. odocoilei*, *B. divergens* and *B. microti*, respectively (Olmeda *et al.* 1997; Armstrong *et al.* 1998). All samples that demonstrated a positive reaction with the PIRO-A and -B primer set were additionally tested with BAB-1 and BAB-4 primers, specific for nss-rDNA of *B. microti* (Persing *et al.* 1992). When using primers PIRO-A and -B, the amplification protocol was slightly changed from the original description; the annealing temperature was raised to 60 °C.

Distinctive amplicons were sequenced on both strands by using the BigDye Terminator Cycle Sequencing Ready Reaction Kit (ABI PRISM, PE Applied Biosystems, California, USA). The sequences obtained were analysed with computer programs SeqMan and EditSeq of the Lasergene 1999 software package (Dnastar, Madison, Winsconsin, USA). Sequence alignments were performed with the MegAlign program of the same software package.

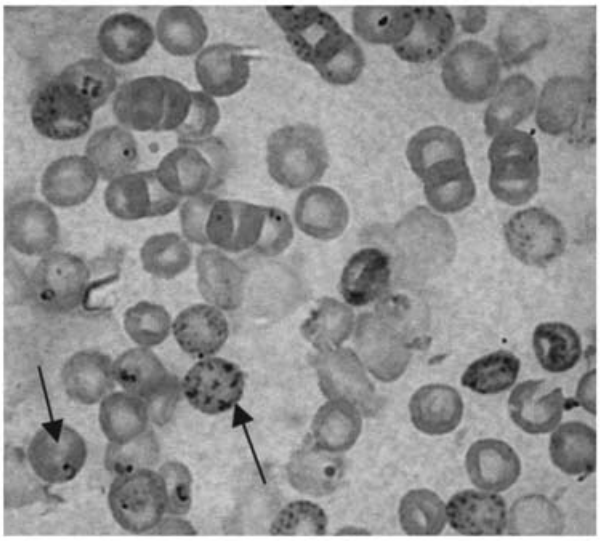


Fig. 1. *Babesia microti* in erythrocytes of *Apodemus flavicollis* trapped in Slovenia. Parasites were mostly shaped as small dots or ring forms, and were located on the periphery of multiply infected erythrocytes.

RESULTS

A total of 261 small mammals comprising 11 species (Table 1) were captured and tested for the presence of babesial parasites. Only *C. glareolus* and *A. flavicollis* were infected with babesiae as detected by PCR (Table 1).

Thin blood smears were not available for all 261 animals tested. However, all animals with positive blood smears were also positive when tested with PCR. Intra-erythrocytic parasites of infected animals displayed various morphologies, mostly small dot and ring forms, which were centrally placed. When the erythrocytes were multiply infected, the parasites were placed on the periphery of the cells (Fig. 1). The characteristic Maltese cross (tetrad) was rarely seen. We also noticed that all animals tested positive with PCR had 8–10 times larger spleen than normally present in mice.

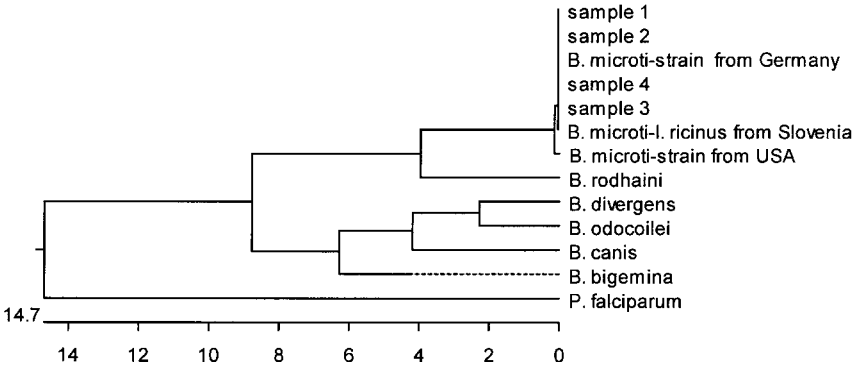


Fig. 2. Phylogenetic relationships of *Babesia* species and tested samples inferred from multiple sequence alignment of 392 bp (primer sequences were removed) of nss-rDNA. All samples were obtained by amplification with PIRO-A and -B primers. Samples 1 and 2 were amplified from *Clethrionomys glareolus*, samples 3 and 4 were amplified from *Apodemus flavicollis*. The sequences obtained were aligned with those of *Babesia divergens* (GenBank accession no. U16370), *B. odocoilei* (U16369), *B. bigemina* (X59607), *B. canis* (L19079), German strain of *B. microti* (AF231349), American strain of *B. microti* (U09833), *B. microti* from tick *Ixodes ricinus* from Slovenia (AF373332), *B. rodhaini* (M87565) and *Plasmodium falciparum* (M19172). Units at the bottom of the phylogenetic tree indicate the percentage of nucleotide substitutions.

When samples were tested with the PIRO-A and -B primer set, 11 (15·9%; 6 males and 5 females) of 69 *C. glareolus* and 15 (11·8%; 12 males and 3 females) of 127 *A. flavicollis*, were infected (Table 1). All samples positive with the PIRO-A and -B primers were additionally positive when tested with primers BAB-1 and -4, specific for *B. microti*. Eight amplicons (4 sized 435 bp and 4 sized 238 bp) from *C. glareolus* and *A. flavicollis* were sequenced. The gene sequences obtained from 435 bp large fragments (PIRO-A and -B primers) showed 100% identity with the *B. microti* strain from Germany (GenBank accession no. AF231349) and 99·7% identity with the *B. microti* strain from USA (U09833) (Fig. 2). Gene sequences sized 238 bp (BAB-1 and -4 primers) were 100% identical to *B. microti* (U09833). The unique sequences determined in this study were deposited in GenBank under the accession nos: AY149572, AY149573.

DISCUSSION

The zoonotic cycle of *B. microti* and its role as an aetiological agent of human babesiosis in the USA is well known (Kjemtrup & Conrad, 2000). In Europe, detection of *B. microti* has been made in a variety of small mammals by microscopical identification in Giemsa's-stained blood smears and by xenodiagnosis. The following 10 species proved to be infected: *C. glareolus*, *Microtus arvalis*, *M. agrestis*, *M. socialis*, *M. nivalis*, *Neomys anomalus*, *A. flavicollis*, *A. agrarius*, *A. sylvaticus* and *Mus musculus* (Sebek, Rosicky & Sixl, 1977; Walter & Liebisch, 1980; Healing, 1981; Karbowiak *et al.* 1999). The prevalence rates for *B. microti* vary between 0·1% and 25% depending on the species and numbers of rodents tested, localities and time of the year (Sebek *et al.* 1977). However, microscopy is labour-intensive and requires well-trained microscopists for accurate identification. Tsuji *et al.* (2001) has demonstrated the use of molecular methods in the case of emerging parasitic infections. They have detected *B. microti* in rodents in Japan, and the parasites were morphologically identical to the American strain of *B. microti* but different from the American strain when sequences of nss-rDNA were compared (Tsuji *et al.* 2001).

In our study, *B. microti* has been detected for the first time in *C. glareolus* and *A. flavicollis* by PCR and subsequent sequence analysis. Although, *A. sylvaticus* and *A. agrarius* were previously implied as hosts of *B. microti* in Europe, no babesial parasites were detected in these 2 animal species collected in Slovenia. In our country, *A. agrarius* is found only in 3 isolated locations and it does not share the same habitat as *A. flavicollis* and *C. glareolus*. However, *A. sylvaticus* is as widely distributed as *A. flavicollis* and *C. glareolus* but it prefers an open habitat, meadows and fields, and it usually harbours less ticks (Krystufek, 1991). The low number of animals per species could be the reason that none of the remaining 7 species were positive. The prevalence rate for *B. microti* determined in our study was 15·9% for *C. glareolus* and 11·8% for *A. flavicollis*. The prevalence rate is higher than in most European countries probably due to the greater sensitivity and specificity of the molecular methods used (Krause *et al.* 1996).

Parasite morphology was, in many cases, similar to the other findings in Europe, yet in some animals the parasites were placed on the periphery of the multiply infected erythrocytes. In a recent study, Gray *et al.* (2002) noticed different morphology in the American and European strain of *B. microti* in infected gerbil erythrocytes. However, the 238 bp long nucleotide sequences of the nss-rDNA of both strains were identical (Gray *et al.* 2002). Small mammals which tested positive for *B. microti* had dramatically enlarged spleens. Similar observations have been

previously described and splenomegaly was indicated as a characteristic feature of *B. microti*-infected animals (Watkins *et al*. 1991). Males were more often infected then females. This is in concordance with other studies and it is ascribed to the greater activity, home range size and high testosterone levels of sexually mature males (Hughes & Randolph, 2001).

The nss-rDNA sequences of *B. microti* from *C. glareolus* and *A. flavicollis* were indistinguishable from each other. Moreover, they were identical with sequences of nss-rDNA of *B. microti* obtained from *I. ricinus* ticks, collected in the same zoogeographical regions in Slovenia (Duh *et al*. 2001). The German strain of *B. microti*, isolated from *I. ricinus* nymphs in Berlin in 1982, was recently sequenced (Zahler, Rinder & Gothe, 2000). When compared, the sequence of *B. microti* strain from Germany (AF231349) showed 100% identity with the sequences of nss-rDNA of *B. microti* obtained from small mammals and from ticks in Slovenia. Only 1 nucleotide difference (G–A, transition mutation) was seen when the above-mentioned sequences were compared to the American strain of *B. microti* (U09833).

The puzzling question of the absence of indigenous human cases of babesiosis caused by *B. microti* in Europe, remains to be answered. The molecular characterization of *B. microti* in small mammals as well as in ticks in Slovenia clearly indicates the wide distribution of this parasite in nature. In Japan, *B. microti*-like parasites were detected in rodents. However, only 1 transfusion-acquired case of human babesiosis has been described so far (Tsuji *et al*. 2001; Wei *et al*. 2001). They suggested that *B. microti* in the USA may be more pathogenic and virulent for humans. Further implication of a lower virulence of the European strain of *B. microti* was recently presented by Foppa *et al*. (2002) in Switzerland. The results of their study indicated that residents in Switzerland are exposed to bites of *I. ricinus* ticks infected with *B. microti*. Namely, 1·5% of human sera were immunoglobulin G-reactive against *B. microti* antigen yet no clinically documented cases of human babesiosis were described (Foppa *et al*. 2002).

In conclusion, our results represent the first study in Europe, in which *B. microti* has been directly identified in *C. glareolus* and *A. flavicollis* by PCR and subsequent sequence analysis.

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